

Geophysical Dynamics of Earth's Rotation, The Deep Water Cycle, and Sea Level Distribution

Introduction

The Earth is not a static, rigid sphere, but rather a dynamic, rotating, viscoelastic ellipsoid governed by an intricate balance of gravitational attraction, centrifugal acceleration, and internal thermodynamics. The distribution of mass across the planet—encompassing the solid lithosphere, the highly viscous mantle, the cryosphere, and the unbound global oceans—is entirely dependent on the conservation of angular momentum and the precise shape of the gravitational equipotential surface known as the geoid. A highly intuitive hypothesis has been proposed suggesting that the Earth's rotational momentum dictates the resting level of the global ocean, functioning similarly to water droplets contained within an inflated, spinning balloon, where centrifugal momentum forces the fluid outward toward the equator.

Furthermore, this hypothesis suggests that historical shifts in the Earth's rotational speed might be fundamentally linked to the onset of massive glacial periods (ice ages), potentially drawing upon newly discovered, vast reservoirs of water located deep within the planet's interior.

To rigorously address these conceptual models, it is necessary to examine the foundational physical mechanisms that dictate the shape of the Earth, the behavior of its oceans, and the composition of its deep interior. The distribution of water on Earth is subject to continual redistribution driven by astronomical forcing, climatic shifts, and deep-time geodynamic cycles. The first core mechanism under review is the interplay between gravity and rotational centrifugal force, which actively sustains the equatorial bulge of both the oceans and the solid Earth. The second mechanism involves the deep water cycle, a geological process by which tectonic subduction and volcanic outgassing exchange water between the surface oceans and the mantle transition zone over millions of years. Finally, the third mechanism encompasses the rapid, climate-driven redistribution of surface mass—specifically the transfer of water between the global oceans and polar ice sheets—and how this continuous redistribution directly alters the planet's moment of inertia, consequently accelerating or decelerating its rotational velocity over time.

This comprehensive report provides an exhaustive synthesis of rotational dynamics, space geodesy, paleoclimatology, and deep-mantle mineralogy. It evaluates the scientific validity of rotational eustasy (sea-level changes driven explicitly by variations in planetary spin), maps the known dimensions and physical states of the mantle's hydrous rock reservoirs, and delineates the true causal pathways between Earth's rotation, the deep water cycle, and the mechanics of global glaciation.

Hydrostatic Equilibrium and Centrifugal Dynamics: The Balloon Analogy

The conceptualization of Earth's surface oceans behaving akin to fluid inside a spinning balloon is highly accurate within the framework of classical geodetic mechanics. Because the Earth is rotating, it is not a perfect sphere; it is an oblate spheroid, flattened at the geographic poles and bulging significantly at the equator.¹ This oblateness is an absolute requirement of hydrostatic equilibrium for any rotating, self-gravitating planetary body.

As the Earth rotates on its axis at an angular velocity (ω) of approximately 7.292115×10^{-5} radians per second, every particle of matter on or within the planet experiences a centrifugal force directed outward, exactly perpendicular to the axis of rotation.² The magnitude of this

centrifugal acceleration (a_c) at the planet's surface is defined by the classical equation $a_c = r\omega^2$, where r is the perpendicular distance from the axis of rotation. At the equator, where the Earth's radius is roughly 6,378,137 meters, this outward centrifugal acceleration

reaches its maximum at approximately 0.03392 m/s^2 .² While this equates to only about 0.34% of the inward gravitational acceleration (which averages 9.81 m/s^2), it is a highly significant force on a planetary scale.³

Because centrifugal force vectors are strictly perpendicular to the axis of rotation, they do not perfectly align with the inward pull of gravity except precisely at the equator.³ In the mid-latitudes, the combination of inward gravity and outward centrifugal force results in a net force vector that slants slightly toward the equator. If the Earth were a completely rigid, perfect sphere, this horizontal force component would cause all oceanic water to flow continuously toward the equator.³ However, because the solid Earth is ductile over geological timescales, the planet itself has deformed into an oblate shape.² The "equatorial bulge" represents a literal hill of matter—encompassing both silicate rock and liquid water—rising toward the equator until the physical slope of the surface is perfectly perpendicular to the effective direction of the modified gravity.³

The surface at which this exact equilibrium is achieved is known as the geoid. The geoid is an equipotential surface of the Earth's gravity field that closely approximates global mean sea level, entirely excluding the localized, transient effects of winds, tides, and ocean currents.⁴ All points on the geoid possess the exact same geopotential, which represents the sum of gravitational potential energy and centrifugal potential energy.⁴ Because the geoid is an equipotential surface, water rests perfectly upon it without flowing equatorward or poleward, satisfying the user's premise that rotational momentum strictly dictates the specific, constant resting level of the oceans during non-ice age periods.³

If the Earth's rotational speed were entirely constant, and if the absolute mass of the oceans remained unchanged, the sea level would indeed be pulled to a highly specific baseline level universally governed by this rotational momentum. The physical dimensions of the equatorial bulge are substantial; the equatorial radius of the Earth is about 21 kilometers greater than its polar radius. If the planet possessed a faster spinning speed in the geological past, the equatorial bulge of oceanic water and the ellipsoidal flattening of the solid Earth were

significantly larger and more pronounced than they are today.¹ The gradual slowing of Earth's rotation over billions of years, primarily driven by tidal friction interactions with the Moon, has caused the solid parts of the Earth and the oceanic bulge to decrease in tandem, confirming that sea level and rotational velocity are permanently coupled.²

Physical Parameter	Value / Formula	Definition and Impact on Ocean Distribution
Angular Velocity (ω)	$7.292115 \times$ rad/s	Earth's rotation rate, driving the centrifugal force that sustains the equatorial ocean bulge. ²
Equatorial Radius (R_E)	6,378,137 m	Distance from the center to the equator; maximizes the r variable in the centrifugal acceleration equation. ²
Centrifugal Acceleration	$\sim$ m/s ²	Outward acceleration at the equator, actively reducing effective gravity and drawing water away from the poles. ³
Dynamic Form Factor (J_2)	$1.08262668 \times$	Coefficient measuring Earth's oblateness; dynamically fluctuates as ice melts or rotation varies over time. ²

Rotational Eustasy: Ocean Level Redistribution in Response to Spin Fluctuations

The hypothesis that a sudden or gradual change in Earth's spin would fundamentally alter ocean levels across the globe is unequivocally supported by geodetic physics, falling under an area of study commonly referred to as "rotational eustasy." Rotational eustasy posits that fluctuations in the Earth's rotational velocity force a massive horizontal redistribution of oceanic water masses between the high latitudes and the equatorial bulge.⁶

Because the oceanic envelope is an unbound, low-viscosity fluid, it responds virtually instantaneously to changes in rotational velocity, whereas the solid lithosphere and viscous

mantle experience a much slower, delayed viscoelastic response.² Therefore, if the Earth were to experience an acceleration in its rotation, centrifugal force would increase, and water would immediately migrate from the polar and mid-latitude regions toward the equator.¹⁰ This dynamic would cause significant relative sea-level rise in equatorial zones (e.g., Brazil, Indonesia, central Africa) and relative sea-level fall at the poles.¹⁰ Conversely, a deceleration in planetary rotation weakens the centrifugal force, causing the massive equatorial bulge to collapse slightly. Gravity would then pull the water poleward, raising sea levels in regions like the Arctic Ocean and the Southern Ocean while dramatically lowering sea levels along the equator.¹

Observational evidence of rotational eustasy has been extensively documented in Holocene geological records. Studies analyzing the coastal morphology, peat deposits, and stratigraphy of equatorial islands (such as the Maldives, Fiji, and New Caledonia) demonstrate cyclical sea-level changes over the past 500 years that diverge significantly from the global glacial-eustatic average.⁶ During the 17th century's Maunder Minimum—a "Grand Solar Minimum" characterized by colder global climates—it is theorized that the Earth experienced a slight rotational acceleration.¹⁰ Stratigraphic evidence from Fiji and the Indian Ocean indicates a +60 to +70 centimeter high sea level during this period, strongly suggesting that water was actively displaced toward the equatorial bulge.⁶

During the warmer 18th century, a corresponding deceleration of rotation reportedly caused sea levels in these exact same equatorial regions to drop well below their present levels as water migrated poleward.⁶ While the precise magnitudes of these historical shifts, and their attribution solely to rotational eustasy, remain subjects of vigorous debate within the broader oceanographic community, the underlying physical mechanism of rotationally-driven ocean redistribution is an established physical reality.⁷ Furthermore, ocean bottom pressure changes resulting from shifts in large-scale ocean circulation patterns can exert torque on the solid Earth, further complicating the feedback loop between the oceans and the length of the day.¹⁴ To fully conceptualize the immense power of rotational momentum over ocean levels, geophysicists model the hypothetical scenario of the Earth's rotation stopping entirely. If the

Earth ceased to spin, the 0.03392 m/s^2 outward centrifugal force at the equator would instantly vanish.³ The oceans, no longer held in the 21-kilometer high equatorial bulge by centrifugal momentum, would migrate rapidly toward the poles to conform to a true sphere dictated solely by Earth's gravity.¹

This massive oceanic migration would completely submerge all of Antarctica, as well as the lowlands of Siberia, northern Europe, and northern Canada, under a vastly deepened Arctic Ocean.¹ Concurrently, the draining of the equatorial bulge would expose massive tracts of the ocean floor, eventually joining the continents into a continuous equatorial megacontinent, separated by two completely distinct polar oceans.¹ This extreme scenario confirms the fundamental validity of the user's initial premise: the exact sea-level lines observed on modern maps are entirely dependent on the specific speed at which the Earth rotates. If the spin changes, the geography of the planet changes.

The Mantle Transition Zone: Earth's "Inner Ocean" of Hydrous Rock

The second major premise of the hypothesis suggests that during ice ages, water is actively pulled from an "inner core ocean" to the surface to form glaciers, and that the true scale of past glaciations can be understood by acknowledging the vast amount of water in the center of the Earth. Modern geophysics confirms that a massive reservoir of water does indeed exist deep within the Earth, but the exact location, physical state, and mechanism of transport differ fundamentally from the concept of a subterranean liquid ocean.

The Earth's core, composed primarily of molten iron and nickel, does not contain massive reservoirs of water, though hydrogen may be dissolved within the metallic iron.¹⁸ However, the mantle transition zone—the highly pressurized boundary layer separating the upper mantle from the lower mantle at depths between 410 and 660 kilometers—contains an immense, globally significant volume of water.¹⁹ This region acts as a massive planetary sponge, theoretically capable of holding up to three to six times the amount of water found in all surface oceans combined.¹⁹

Crucially, this deep-earth water does not exist in a liquid, gaseous, or solid ice state, meaning there is no sloshing ocean to be pumped to the surface. Instead, the water is chemically bound within the crystalline structure of ultra-high-pressure hydrous rocks.¹⁹ The Earth's upper mantle is predominantly composed of the silicate mineral olivine. As depth increases, extreme lithostatic pressure (reaching up to 23,000 bar) forces the olivine to adopt much denser crystalline structures.¹⁹ At a depth of 410 kilometers, olivine converts into a mineral called wadsleyite, and at 520 kilometers, it metamorphoses into an even denser phase known as ringwoodite.¹⁹

Unlike standard olivine, which is nominally anhydrous, the unique crystalline lattices of wadsleyite and ringwoodite have an enormous capacity to store water in the form of trapped hydroxyl ions (OH⁻).²⁰ The existence of this "wet" transition zone was definitively proven in 2014, and spectacularly reaffirmed in 2022, when geologists discovered ultra-deep, super-rare diamonds expelled to the surface via violent volcanic diatreme eruptions.²¹ A 1.5-centimeter diamond retrieved from Botswana, formed exactly at the 660-kilometer boundary, contained microscopic inclusions of ringwoodite with an exceptionally high water content, proving beyond a doubt that the deep mantle is highly hydrated.¹⁹

If a large boulder of this ringwoodite were somehow brought intact to the surface, it would neither feel wet nor drip water.¹⁹ However, when subjected to "dehydration melting"—a process where the rock is forced to melt due to changes in pressure or temperature—it releases oceanic volumes of liquid H₂O.¹⁹ This is verified by seismologists mapping the planet's interior; seismic waves generated by over 500 earthquakes were observed slowing down significantly as they passed through the transition zone beneath North America.²⁰ Because acoustic waves take longer to travel through hydrated rock and partial melt than through dry rock, this seismic velocity anomaly provided the macroscopic proof of the transition zone's vast water reserves.²⁰ While it brings science one step closer to Jules Verne's idea of an ocean

inside the Earth, the reality is one of solid, glowing-hot, hydrated rock.¹⁹

Mantle Layer / Mineral	Depth	Water Storage Mechanism	Hydration Capacity
Upper Mantle (Olivine)	0 to 410 km	Nominally anhydrous	Very Low (50–200 ppm) ²⁶
Transition Zone (Wadsleyite)	410 to 520 km	Hydroxyl ions (OH ⁻) in lattice	High (Up to 2% by weight) ¹⁹
Transition Zone (Ringwoodite)	520 to 660 km	Hydroxyl ions (OH ⁻) in lattice	High (Up to 2% by weight) ¹⁹
Lower Mantle (Bridgmanite)	660 to 2,900 km	Dense phases, highly pressurized	Low (water expelled at 660km boundary) ¹⁸

The Deep Water Cycle vs. Rapid Glacial Oscillations

The transport of water between the surface oceans and the transition zone is governed by a planetary-scale process known as the "Deep Water Cycle".²⁶ Tectonic subduction acts as a massive planetary conveyor belt. As cold, dense oceanic plates collide and sink beneath lighter continental plates at subduction zones, they drag hydrated minerals, deep-sea sediments, and trapped oceanic water downward deep into the mantle—a process known as regassing.¹⁹

Conversely, water is slowly returned to the surface—degassing—through volcanic activity at mid-ocean ridges, volcanic island arcs, and deep mantle plumes (hotspots like Hawaii).²⁶

While the transition zone houses vast amounts of water, it is physically impossible for this deep-earth reservoir to act as the primary source of water for surface glaciation during ice ages, which refutes a portion of the user's core hypothesis. The limiting factor is the extreme disparity in mass flux and temporal scales.

The global flux of outgassing water through mid-ocean ridge volcanism and arc volcanism is

estimated to be approximately 6×10^{10} to 1.2×10^{11} kilograms per year.²⁹ At this steady outgassing rate, it would take roughly 12 billion years to "make" or fill the modern global ocean.³⁰ The residence time for water locked in the deep mantle is measured in hundreds of millions to billions of years, whereas the water within the surface hydrological cycle (oceans, atmosphere, glaciers) has a highly rapid turnover.³¹

Ice ages, specifically those of the Pleistocene epoch within the broader Late Cenozoic Icehouse state, operate on relatively rapid cycles of 20,000 to 100,000 years.³³ This cyclic timeframe is astronomically too fast to be driven by the slow, grinding volcanic outgassing from the deep mantle. If mantle water were venting rapidly enough to raise global sea levels and

provide enough precipitation to drop sea levels by 134 meters during an ice age, the associated volcanic and seismic activity required to expel that water would be cataclysmic beyond anything recorded in geological history, likely sterilizing the surface. Therefore, while the Earth absolutely contains an inner reservoir holding oceans' worth of water, this water remains securely trapped in the crystal lattices of ringwoodite and is largely isolated from the rapid, climate-driven fluctuations of the surface ice ages.

Orbital Forcing, Mass Redistribution, and the Mechanics of the Cryosphere

If water is not venting from the inner Earth to create glaciers, where does the ice come from, and how does it relate to the Earth's changing spin?

The true origin of ice age water is the surface ocean itself. The Earth's climate naturally alternates between greenhouse and icehouse periods. The current Late Cenozoic Ice Age began roughly 34 million years ago, characterized by long-term cooling trends and the persistent presence of polar ice sheets.³⁴ Within this long-term icehouse framework, the planet experiences rapid, severe oscillations between intense glacial periods and mild interglacial periods (like the current Holocene epoch, which began roughly 11,700 years ago).³³

These glacial cycles are not driven by the deep mantle, but rather by extraterrestrial forcing—specifically astronomical phenomena known as "Milankovitch cycles." These are predictable, cyclical variations in the Earth's orbital eccentricity (the elliptical shape of its orbit around the sun), its axial tilt (obliquity), and its precession (axial wobble).³³ When these orbital parameters align to significantly reduce summer solar radiation in the Northern Hemisphere, snow that falls during the winter fails to melt during the summer.³³

Over tens of thousands of years, water continuously evaporates from the warm oceans, travels through the atmosphere as water vapor, and falls as snow on landmasses like North America, northern Europe, and Antarctica.³⁶ Because the snow does not melt, it continually accumulates layer by layer, compressing under its own immense weight into massive continental ice sheets up to three miles thick.³⁶ This atmospheric process actively pulls water out of the oceans, causing a rapid and dramatic global fall in sea level. During the Last Glacial Maximum (approximately 26,000 to 20,000 years ago), the withdrawal of ocean water to form these massive ice sheets caused global sea levels to plummet to approximately 134 meters below their current level.³⁷ The grounded ice volume during this period was roughly 52 million cubic kilometers greater than it is today.³⁷

Conservation of Angular Momentum and Glacial Isostatic Adjustment

This massive transfer of water from the equatorial and mid-latitude oceans to the high-latitude polar ice caps fundamentally alters the Earth's moment of inertia, which in turn directly alters its rotational speed.

The physical law governing this massive planetary process is the Conservation of Angular

Momentum. A system's angular momentum (L) is the product of its moment of inertia (I)

and its angular velocity (ω), represented mathematically by the equation $L = I\omega$.³⁹ In an isolated system without external torques, total angular momentum must remain constant. The

moment of inertia (I) is a precise measure of how a body's mass is distributed relative to its axis of rotation; mass located further from the axis (at the equator) contributes exponentially more to the moment of inertia than mass located near the axis (at the poles).⁴¹

During the onset of an ice age, billions of tons of water evaporate from the equatorial oceans (far from the axis of rotation) and are eventually deposited as highly compressed ice at the poles (very close to the axis of rotation).³⁶ Because an enormous amount of mass is shifting

closer to the rotational axis, the Earth's total moment of inertia (I) decreases.⁴⁰ In order to

strictly conserve angular momentum (L), the angular velocity (ω) must increase.

Consequently, the massive formation of glaciers actually causes the Earth to spin faster, slightly shortening the length of the planetary day.⁵

The Role of Glacial Isostatic Adjustment (GIA)

The relationship between glaciation and spin is further complicated by the response of the solid Earth beneath the ice, a geodynamic process known as Glacial Isostatic Adjustment (GIA) or post-glacial rebound.⁹ The immense, crushing weight of continental ice sheets depresses the solid lithosphere, forcing the highly viscous mantle material beneath it to flow away from the poles and toward the equator.⁹ This sub-surface process shifts solid mass away from the axis of rotation, which partially counteracts the decrease in the moment of inertia caused by the surface water movement.

When the ice age ends and the glaciers melt, the meltwater rapidly returns to the oceans, increasing the moment of inertia and slowing the Earth's spin.⁴¹ However, the depressed continental crust takes tens of thousands of years to "rebound" because the underlying mantle rock is highly viscous and flows extremely slowly (like thick syrup).⁹ Because the Earth is still actively undergoing post-glacial rebound from the end of the Last Glacial Maximum 11,700 years ago, mantle mass is currently still migrating deep underground from the equatorial regions back toward the poles to restore equilibrium.⁴⁷

This ongoing upward rebound of landmasses (most notably in Canada and Scandinavia) is

continuously reducing the Earth's oblateness (the J_2 coefficient) and slowly decreasing the moment of inertia.² As a direct result of this specific post-glacial GIA process, the Earth's rotation has a long-term geological tendency to slowly accelerate, shortening the day by a fraction of a millisecond per century.⁵ Therefore, the legacy of the last ice age is literally imprinted on the Earth's rotational momentum today.

Furthermore, GIA induces "true polar wander," a phenomenon where the redistribution of mantle mass alters the principal axes of the planet's inertia tensor.³⁹ This causes the entire solid body of the Earth to slowly rotate relative to its spin axis.⁵¹ The rotational fluctuations driven by GIA also exert a profound feedback effect on the oceans; as the rotation vector shifts, the

centrifugal potential across the planet is slightly modified, inducing spatial variations in the geoid and causing long-term, localized sea-level changes completely distinct from the sheer volumetric addition of meltwater.⁹

Contemporary Climate Change and Unprecedented Rotational Deceleration

While the long-term baseline of Earth's rotation is influenced by lunar tidal friction (which slows the planet) and Glacial Isostatic Adjustment (which speeds the planet up), the modern era has introduced a rapid, overwhelmingly dominant variable: anthropogenic climate change.

Observations of modern sea-level rise provide direct, contemporary validation of the hypotheses regarding mass distribution, centrifugal forces, and spin. Since 1960, global sea levels have risen by an average of 2.06 millimeters per year, a rate that has sharply accelerated to 3.94 millimeters per year between 2005 and 2023.⁵² This rapid sea-level rise is driven by two primary factors: the thermal expansion of seawater as it absorbs atmospheric heat, and the accelerated melting of the Greenland and Antarctic ice sheets, alongside mountain glaciers.⁵² Ocean warming accounts for approximately 43% of the total sea-level rise since 1960, while melting mountain glaciers (27%), the Greenland Ice Sheet (15%), and the Antarctic Ice Sheet (12%) constitute the bulk of the remaining added water mass.⁵²

This rapid introduction of meltwater is enacting the exact inverse of the glaciation physics described above. As massive volumes of polar ice melt, the resulting liquid water flows into the global oceans and spreads laterally toward the equator to conform to the planet's geoid.⁵⁴ This barystatic process redistributes mass from the poles (close to the axis of rotation) to the equatorial ocean bulge (far from the axis of rotation).⁵⁵ In a perfect illustration of the conservation of angular momentum—often likened by physicists to a figure skater extending their arms outward during a spin—this equatorial mass loading increases the planet's moment of inertia and forces a rapid deceleration of the Earth's rotation.⁴²

Advanced geodetic analyses, utilizing deep-learning artificial intelligence algorithms applied to 120 years of astronomical and satellite data, confirm that 90% of the recurring fluctuations in polar motion and rotational deceleration over the last century can be explained by changes in ice sheets, glaciers, sea levels, and severe groundwater depletion.⁵¹ Modern space geodesy techniques, including Very Long Baseline Interferometry (VLBI) and Global Navigation Satellite Systems (GNSS), indicate that the mass melting of polar ice is currently slowing the planet's rotation at an astonishing rate of roughly 1.33 milliseconds per century.⁵

While 1.33 milliseconds may appear imperceptible to humans, it is a massive figure in geophysical terms. Paleoclimatic data, derived from microscopic marine fossils and sediment cores, reveal that the current rate of day-length increase is completely unprecedented over the past 3.6 million years (stretching back to the late Pliocene epoch).⁴¹ The rapid warming of the planet is actively overpowering the long-term rotational acceleration typically provided by post-glacial rebound, forcing the planet into a decelerating state.⁴³

This rotational deceleration also introduces highly complex secondary effects on ocean levels that must be accounted for in precision global models. As the Earth's rotation slows, the

centrifugal force at the equator weakens slightly.³ This requires highly localized, constant recomputations of the geoid for precise orbit determination (POD) of altimetry satellites like Jason-1 and TOPEX/Poseidon.⁶⁰ A failure to account for these millisecond changes in Length of Day (LOD) would introduce millimeter-level errors in global mean sea level estimations, corrupting climate data.⁶⁰ Furthermore, mathematical models extrapolated to the year 2050 suggest that if long-period rotational oscillations (such as recently discovered 64-year cycles) and multi-decadal sea-level feedback loops are ignored, predictive errors in sea-level rise trajectories could exceed 11%.⁶²

Conclusion

The distribution of the world's oceans is the result of a highly delicate, continually adjusting equilibrium between the inward pull of gravity, outward centrifugal dynamics, orbital geometry, and the deep-earth material structure. Based on an exhaustive review of geophysics, geodesy, paleoclimatology, and deep mantle mineralogy, several distinct conclusions can be drawn regarding the mechanics of Earth's rotation, its interior water, and its climate states.

First, the core hypothesis that Earth's rotation pulls water to a specific, constant resting level during stable climatic periods is physically accurate. The centrifugal force generated by the Earth's spin directly counteracts gravity, resulting in an equatorial bulge that elevates the solid earth and the sea surface approximately 21 kilometers higher at the equator than at the poles. The geoid—the equipotential surface representing mean sea level—is fundamentally defined by this rotational momentum. A permanent change in rotational speed absolutely dictates a massive change in global sea-level distribution, with a deceleration causing water to migrate poleward, and an acceleration drawing water to the equator (a process known as rotational eustasy).

Second, while the presence of a vast, inner-Earth water reservoir is confirmed by modern seismology, the hypothesis that it acts as the immediate source of water for surface glaciation is unsupported by tectonic timescales and mass flux data. The mantle transition zone, spanning depths of 410 to 660 kilometers, contains immense quantities of water trapped as hydroxyl ions within the crystal structures of wadsleyite and ringwoodite minerals. However, the deep water cycle operates on a scale of tens of millions to billions of years via tectonic subduction and volcanic outgassing. It is far too slow to drive the rapid, tens-of-thousands-of-years cycles that govern the Pleistocene ice ages.

Instead, ice ages are initiated by cyclical variations in Earth's orbit (Milankovitch cycles), which reduce solar radiation and prevent summer ice melt in the high latitudes. The water forming these massive, three-mile-thick continental glaciers is sourced directly from the evaporation of the surface oceans, causing dramatic sea-level drops exceeding 134 meters during a glacial maximum.

Finally, the relationship between sea level, glaciation, and Earth's spin is inextricably linked through the conservation of angular momentum. The formation of polar ice sheets shifts mass inward toward the axis of rotation, decreasing the moment of inertia and speeding up the Earth's rotation. Conversely, the contemporary melting of the Greenland and Antarctic ice sheets due to anthropogenic climate change is rapidly transferring mass back into the equatorial oceans. This barystatic redistribution is increasing the Earth's moment of inertia,

slowing its rotation, and lengthening the day at a rate of 1.33 milliseconds per century—a speed unprecedented in the last 3.6 million years.

Ultimately, the Earth is not a rigid sphere, but a highly fluid, viscoelastic system. Its oceans operate exactly like a liquid responding to the centrifugal limits of a spinning vessel, forever adjusting to the shifting parameters of climate, crustal deformation, and the profound, inescapable physics of planetary momentum.

Works cited

1. If the Earth Stood Still: Modeling the Absence of Centrifugal Force - Esri, accessed May 30, 2026, <https://www.esri.com/about/newsroom/arcuser/what-would-happen-if-earth-stopped-spinning>
2. Equatorial bulge - Wikipedia, accessed May 30, 2026, https://en.wikipedia.org/wiki/Equatorial_bulge
3. 24a.The Rotating Earth - NASA, accessed May 30, 2026, <https://pwg.gsfc.nasa.gov/stargaze/Srotfram1.htm>
4. Geoid - Wikipedia, accessed May 30, 2026, <https://en.wikipedia.org/wiki/Geoid>
5. The increasingly dominant role of climate change on length of day variations - PNAS, accessed May 30, 2026, <https://www.pnas.org/doi/10.1073/pnas.2406930121>
6. Rotational Eustasy as Observed in Nature - Scirp.org., accessed May 30, 2026, <https://www.scirp.org/journal/paperinformation?paperid=94056>
7. (Open Access) Our Oceans–Our Future: New Evidence-based Sea Level Records from the Fiji Islands for the Last 500 years Indicating Rotational Eustasy and Absence of a Present Rise in Sea Level (2017) | Nils-Axel Mörner | 10 Citations - SciSpace, accessed May 30, 2026, <https://scispace.com/papers/our-oceans-our-future-new-evidence-based-sea-level-records-3a1exf48f1>
8. New Evidence-based Sea Level Records from the Fiji Islands for the Last 500 years Indicating Rotational E, accessed May 30, 2026, <https://pdfs.semanticscholar.org/86c0/c76de92a5b097735a7a186fdf767a6124ab8.pdf>
9. Glacial isostatic adjustment (GIA), accessed May 30, 2026, <https://www.gfz.de/en/section/earth-system-modelling/topics/solid-earth-dynamics/glacial-isostatic-adjustment-gia>
10. Rotational Eustasy as Understood in Physics - SCIRP, accessed May 30, 2026, <https://www.scirp.org/journal/paperinformation?paperid=93396>
11. Our Oceans–Our Future: New Evidence-based Sea Level Records from the Fiji Islands for the Last 500 years Indicating Rotational Eustasy and Absence of a Present Rise in Sea Level - Graphy Publications, accessed May 30, 2026, <https://www.graphyonline.com/archives/IJEES/2017/IJEES-137/>
12. Absolute Evidence of the Absence of an on-Going Sea Level Rise on Ouvéa Island of New Caledonia - SSRG International Journals, accessed May 30, 2026, <https://www.internationaljournalsssrg.org/IJGGS/2018/Volume5-Issue3/IJGGS-V5I>

[3P104.pdf](#)

13. (PDF) Rotational Eustasy as Understood in Physics - ResearchGate, accessed May 30, 2026,
https://www.researchgate.net/publication/334135324_Rotational_Eustasy_as_Understood_in_Physics
14. A Consistent Sea-Level Reconstruction and Its Budget on Basin and Global Scales over 1958–2014 in - AMS Journals, accessed May 30, 2026,
<https://journals.ametsoc.org/view/journals/clim/31/3/jcli-d-17-0502.1.xml>
15. Trends and interannual variability of mass and steric sea level in the Tropical Asian Seas - IIASA PURE, accessed May 30, 2026,
<https://pure.iiasa.ac.at/id/eprint/14739/1/jgrc22395.pdf>
16. If Earth Stopped Spinning - YouTube, accessed May 30, 2026,
https://www.youtube.com/shorts/kzHAeFY_TEM
17. What Would Happen If the Earth Stopped Spinning? | UC Geography, accessed May 30, 2026,
<https://legacy.geog.ucsb.edu/what-would-happen-if-the-earth-stopped-spinning/>
18. The role of water in Earth's mantle - PMC - NIH, accessed May 30, 2026,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8288861/>
19. An ocean inside the Earth? Water hundreds of kilometres down ..., accessed May 30, 2026,
<https://aktuelles.uni-frankfurt.de/en/english/an-ocean-inside-the-earth-water-hundreds-of-kilometres-down/>
20. Massive 'ocean' discovered in Earth's mantle - Geological Society of Glasgow, accessed May 30, 2026,
<https://geologyglasgow.org.uk/headlines/massive-ocean-discovered-in-earths-mantle/>
21. Ringwoodite - Wikipedia, accessed May 30, 2026,
<https://en.wikipedia.org/wiki/Ringwoodite>
22. An Ocean inside the Earth? Water Hundreds of Kilometers Down | Lab Manager, accessed May 30, 2026,
<https://www.labmanager.com/an-ocean-inside-the-earth-water-hundreds-of-kilometers-down-28869>
23. Water distribution on Earth - Wikipedia, accessed May 30, 2026,
https://en.wikipedia.org/wiki/Water_distribution_on_Earth
24. An Underground Ocean? Scientists Discover Water Deep Within Earth - SciTechDaily, accessed May 30, 2026,
<https://scitechdaily.com/an-underground-ocean-scientists-discover-water-deep-within-earth/>
25. New Evidence for Oceans of Water Deep in the Earth | BNL Newsroom - Brookhaven National Laboratory, accessed May 30, 2026,
<https://www.bnl.gov/newsroom/news.php?a=111648>
26. Water in Earth's mantle - Physics Today, accessed May 30, 2026,
<https://physicstoday.aip.org/features/water-in-earths-mantle>
27. Deep water cycle - Wikipedia, accessed May 30, 2026,

- https://en.wikipedia.org/wiki/Deep_water_cycle
28. Water and Mantle Dynamics - Clint Conrad, accessed May 30, 2026, https://www.clintconrad.no/waterplanet/WP_mantle.html
 29. Distribution, cycling and impact of water in the Earth's interior - Oxford Academic, accessed May 30, 2026, <https://academic.oup.com/nsr/article/4/6/879/4569204>
 30. The Geological History of Water: From Earth's Accretion to the Modern Deep Water Cycle | Elements | GeoScienceWorld, accessed May 30, 2026, <https://pubs.geoscienceworld.org/msa/elements/article/20/4/253/646228/The-Geological-History-of-Water-From-Earth-s>
 31. Water Versus Land on Temperate Rocky Planets - PMC - NIH, accessed May 30, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12783251/>
 32. Water cycle - Wikipedia, accessed May 30, 2026, https://en.wikipedia.org/wiki/Water_cycle
 33. Scientists match Earth's ice age cycles with orbital shifts | The Current - UCSB, accessed May 30, 2026, <https://news.ucsb.edu/2025/021777/scientists-match-earths-ice-age-cycles-orbital-shifts>
 34. Ice age - Wikipedia, accessed May 30, 2026, https://en.wikipedia.org/wiki/Ice_age
 35. Deep-Water Sediments Reveal Cyclical Patterns of Extraterrestrial Influence on Earth's Ancient and Modern Climates | lettersandsciencemag, accessed May 30, 2026, <https://lettersandsciencemag.ucdavis.edu/science-technology/deep-water-sediments-reveal-cyclical-patterns-extraterrestrial-influence-earths>
 36. Ice, Snow, and Glaciers and the Water Cycle | U.S. Geological Survey - USGS.gov, accessed May 30, 2026, <https://www.usgs.gov/water-science-school/science/ice-snow-and-glaciers-and-water-cycle>
 37. Sea level and global ice volumes from the Last Glacial Maximum to the Holocene - PMC, accessed May 30, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4217469/>
 38. December | 2017 | Karst Geochemistry and Hydrogeology - NUSites, accessed May 30, 2026, <https://sites.northwestern.edu/monroyrios/2017/12/>
 39. Global Glacial Isostatic Adjustment: Target Fields for Space Geodesy, accessed May 30, 2026, <https://ilrs.gsfc.nasa.gov/lw15/docs/presents/Monday1440.pdf>
 40. Postglacial rebound and sea level contributions to changes in the geoid and the Earth's rotation axis | Geophysical Journal International | Oxford Academic, accessed May 30, 2026, <https://academic.oup.com/gji/article/136/3/537/651433>
 41. Climate change is making days longer, study says - Astronomy Magazine, accessed May 30, 2026, <https://www.astronomy.com/science/climate-change-is-making-days-longer-study-says/>
 42. Why is climate change triggering faster rotation? - Physics Stack Exchange, accessed May 30, 2026, <https://physics.stackexchange.com/questions/721809/why-is-climate-change-trig>

[gering-faster-rotation](#)

43. Will the day/night cycle change when the poles melt? - Physics Stack Exchange, accessed May 30, 2026, <https://physics.stackexchange.com/questions/176429/will-the-day-night-cycle-change-when-the-poles-melt>
44. Glacial isostatic adjustment and sea-level change – State of the art report - SKB, accessed May 30, 2026, <https://skb.se/upload/publications/pdf/TR-09-11.pdf>
45. The rotational feedback on linear-momentum balance in glacial isostatic adjustment - Oxford Academic, accessed May 30, 2026, <https://academic.oup.com/gji/article/199/3/1823/620848>
46. Isostasy 2 'Bouncing back' after the ice - Earth Learning Idea, accessed May 30, 2026, https://www.earthlearningidea.com/PDF/206_Isostasy_2.pdf
47. Earth's rotation - Wikipedia, accessed May 30, 2026, https://en.wikipedia.org/wiki/Earth%27s_rotation
48. W.R. Peltier, R.A. Drummond and A.M. Tushingham, Post-Glacial Rebound and Transient Lower Mantle Rheology, Geophys. J.R. Astr. Soc., 87, 79-116, 1986 - Earth, Atmospheric, and Planetary Physics, accessed May 30, 2026, https://www.atmosp.physics.utoronto.ca/~peltier/pubs_highestimpact/W.R.%20Peltier,%20R.A.%20Drummond%20and%20A.M.%20Tushingham,%20Post-Glacial%20Rebound%20and%20Transient%20Lower%20Mantle%20Rheology,%20Geophys.%20J.R.%20Astr.%20Soc..%2087.%2079-116.%201986.pdf
49. Can we see evidence of post-glacial geoidal adjustment in the current slowing rate of rotation of the Earth?, accessed May 30, 2026, https://www.eps.mcgill.ca/~courses/c510/Barreto_Fortin_Iredale.pdf
50. Glacial isostatic adjustment on a rotating earth - NASA ADS, accessed May 30, 2026, <https://ui.adsabs.harvard.edu/abs/2001GeoJI.147..562M>
51. NASA-Funded Studies Explain How Climate Is Changing Earth's Rotation, accessed May 30, 2026, <https://www.nasa.gov/science-research/earth-science/nasa-funded-studies-explain-how-climate-is-changing-earths-rotation/>
52. Scientists Just Confirmed What's Driving Sea Level Rise And It's Alarming - SciTechDaily, accessed May 30, 2026, <https://scitechdaily.com/scientists-just-confirmed-whats-driving-sea-level-rise-and-its-alarming/>
53. Temporal changes to the geoid and vertical datum - National Geodetic Survey (NGS), accessed May 30, 2026, https://www.ngs.noaa.gov/GRAV-D/2016SummerSchool/presentations/day-5/1DruSmith_TempGeoid.pdf
54. Something 'unprecedented' is now happening to Earth's rotation, scientists say, accessed May 30, 2026, <https://www.sciencefocus.com/news/unprecedented-earths-rotation>
55. New perspectives on increase in the length of day caused by climate change, accessed May 30, 2026, <https://communities.springernature.com/posts/new-perspectives-on-increase-in-the-length-of-day-caused-by-climate-change>

56. How climate change is altering the Earth's rotation, accessed May 30, 2026, <https://www.gfz.de/en/press/news/details/wie-der-klimawandel-die-erdrotation-veraendert>
57. accessed May 30, 2026, <https://www.astronomy.com/science/climate-change-is-making-days-longer-study-says/#:~:text=Thanks%20to%20climate%20change%20and,about%201.33%20milliseconds%20per%20century>
58. Climate change slows Earth's spin: Day lengthening unprecedented in 3.6 million years - Universität Wien, accessed May 30, 2026, <https://www.univie.ac.at/en/news/detail/climate-change-slows-earths-spin-day-lengthening-unprecedented-in-36-million-years>
59. Climate change slows Earth's spin: Day lengthening unprecedented in 3.6 million years, accessed May 30, 2026, <https://www.eurekalert.org/news-releases/1119694>
60. The Precision Behind Sea Level Rise | NASA Earthdata, accessed May 30, 2026, <https://www.earthdata.nasa.gov/news/feature-articles/precision-behind-sea-level-rise>
61. updated - Climate4you, accessed May 30, 2026, http://www.climate4you.com/Text/Climate4you_October_2017.pdf
62. Full article: Multi-HSA detection of long-period sea-level variability: application to the quasi-60-yr geodynamic oscillation - Taylor & Francis, accessed May 30, 2026, <https://www.tandfonline.com/doi/full/10.1080/10095020.2025.2578578>